

Nyquist Stability Criterion

Closed-loop stability from encirclements of -1 .

Ogata, *Modern Control Engineering*, Ch. 7.

In this tutorial you will:

- apply $Z = N + P$ to read closed-loop stability off a Nyquist plot,
- work a stable-open-loop and an unstable-open-loop ($P > 0$) example, and
- watch a stable design lose stability as the gain crosses K_{cr} .

Step through with **Ctrl+Enter**, or render a report with **publish**.

For a unity-feedback loop with open-loop transfer function $G(s)H(s)$, the Nyquist criterion relates the number of closed-loop right-half-plane (RHP) poles Z to the number of open-loop RHP poles P and the number of clockwise encirclements N of the point $-1 + j0$ made by the Nyquist plot of $G(j\omega)H(j\omega)$ as ω traverses the Nyquist contour:

$$Z = N + P$$

The closed-loop system is stable iff $Z = 0$, i.e. $N = -P$ (the locus must encircle -1 counterclockwise exactly P times).

Contents

- Worked Example: Stable Open Loop, Determine Closed-Loop Stability
- Critical Gain from the Nyquist/Routh Boundary
- Worked Example: Unstable Open-Loop Plant ($P > 0$)
- Before vs. After: Crossing the Critical Gain
- Try it yourself

Worked Example: Stable Open Loop, Determine Closed-Loop Stability

$G(s) = \frac{K}{(s+1)(s+2)(s+3)}$. The open-loop system has $P = 0$ RHP poles (all open-loop poles are in the LHP), so for closed-loop stability we need $N = 0$ – the Nyquist plot must NOT encircle -1 .

```
K = 60;
G = tf(K,conv(conv([1 1],[1 2]),[1 3]));
fprintf('Open-loop poles:\n'); disp(pole(G))

figure
nyquist(G)
title('Nyquist Plot: $G(s)=\frac{60}{(s+1)(s+2)(s+3)}$', 'Interpreter', 'latex', 'FontSize', 20)
grid on

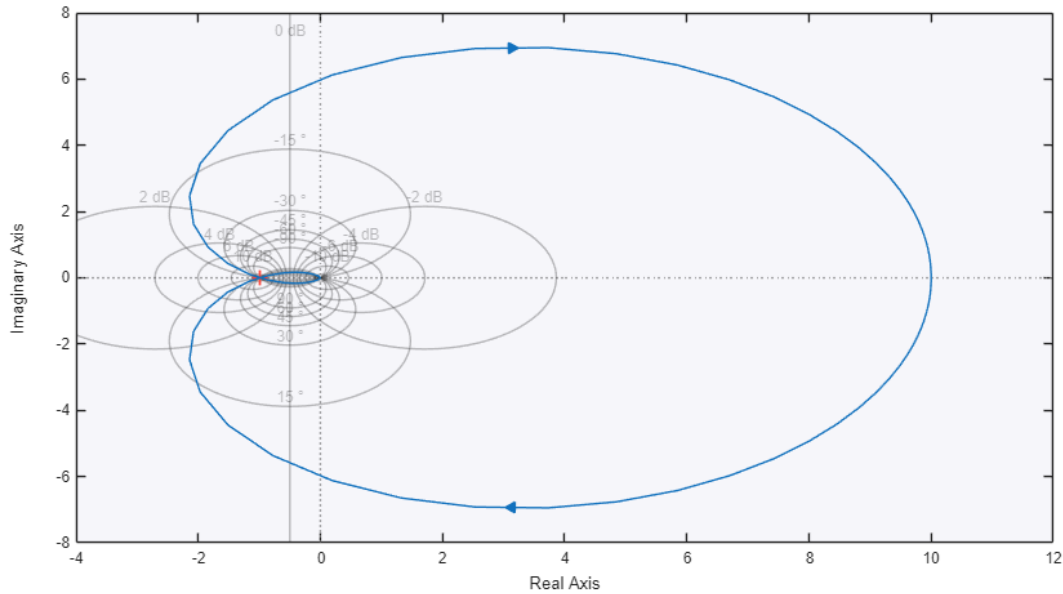
T = feedback(G,1);
cl_poles = pole(T);
fprintf('Closed-loop poles at K=%d:\n', K); disp(cl_poles)
n_rhp = sum(real(cl_poles)>0);
fprintf('Number of closed-loop RHP poles (Z) = %d\n', n_rhp)

Open-loop poles:
-3.0000
-2.0000
-1.0000

Closed-loop poles at K=60:
-6.0000 + 0.0000i
-0.0000 + 3.3166i
-0.0000 - 3.3166i

Number of closed-loop RHP poles (Z) = 0
```

$$\text{Nyquist Plot: } G(s) = \frac{60}{(s+1)(s+2)(s+3)}$$



Critical Gain from the Nyquist/Routh Boundary

The characteristic equation is $(s+1)(s+2)(s+3) + K = 0$, i.e. $s^3 + 6s^2 + 11s + (6+K) = 0$. By Routh's array (see `Transient and Steady-State/RouthStability.m`), stability requires $0 < K < 60$ – so $K = 60$ above is exactly the marginal case, and the Nyquist plot should pass through $-1 + j0$.

```
Kcr = 60;
fprintf('Predicted critical gain Kcr = %d (locus passes through -1)\n', Kcr)
```

```
Predicted critical gain Kcr = 60 (locus passes through -1)
```

Worked Example: Unstable Open-Loop Plant ($P > 0$)

$G(s) = \frac{K}{s(s-1)}$ has one open-loop RHP pole ($P = 1$). For closed-loop stability we now need $N = -1$, i.e. exactly one **counterclockwise** encirclement of -1 .

```
G3 = tf(1,[1 -1 0]);
fprintf('Open-loop poles of G3:\n'); disp(pole(G3))

figure
nyquist(G3)
title('Nyquist Plot: $G(s)=\frac{1}{s(s-1)}$ (Unstable Open Loop, P=1)', 'Interpreter','latex', 'FontSize',20)
grid on
```

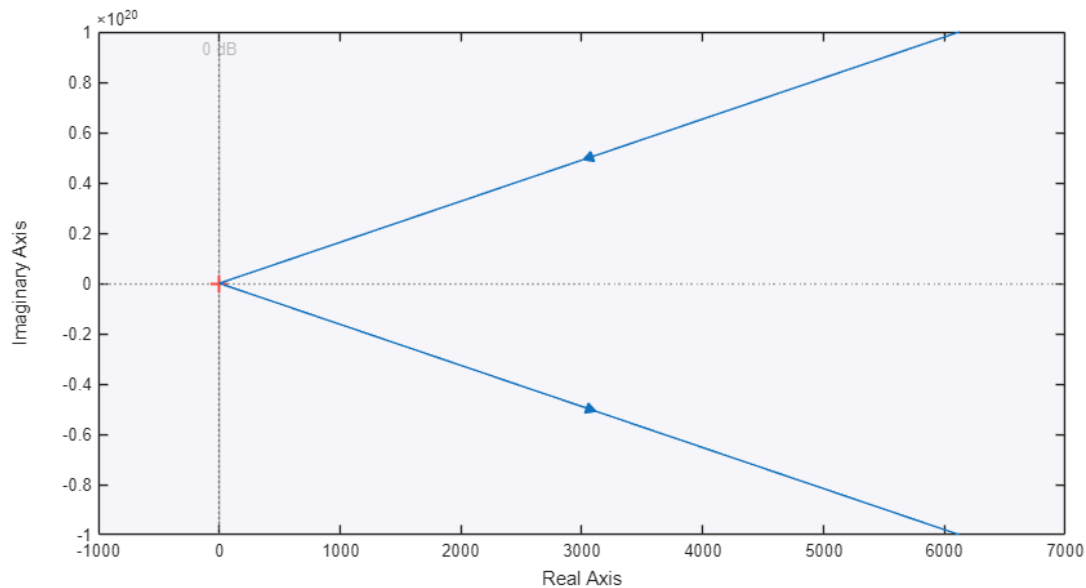
```
T3 = feedback(G3,1);
fprintf('Closed-loop poles of G3/(1+G3):\n'); disp(pole(T3))
fprintf('Closed-loop stable: %d\n', all(real(pole(T3))<0))
```

```
Open-loop poles of G3:
0
1
```

```
Closed-loop poles of G3/(1+G3):
0.5000 + 0.8660i
0.5000 - 0.8660i
```

```
Closed-loop stable: 0
```

Nyquist Plot: $G(s) = \frac{1}{s(s-1)}$ (Unstable Open Loop, P=1)

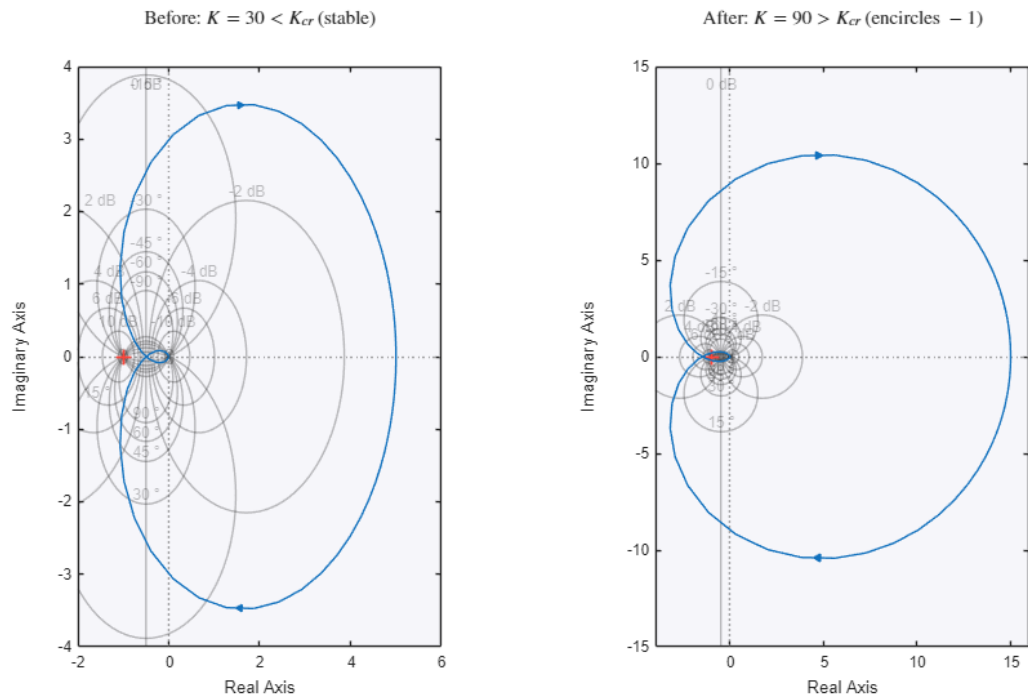


Before vs. After: Crossing the Critical Gain

Back to the first plant. Below K_{cr} the Nyquist plot stays to the right of -1 (no encirclement, closed-loop stable); above K_{cr} it encircles -1 ($Z = 2$, unstable). This is the same stability boundary the root locus crosses at $K = K_{cr}$.

```
den3 = conv(conv([1 1],[1 2]),[1 3]);
figure
subplot(1,2,1)
nyquist(tf(30,den3))
title('Before: $K=30<K_{cr}$ (stable)', 'Interpreter', 'latex', 'FontSize', 13)
grid on
subplot(1,2,2)
nyquist(tf(90,den3))
title('After: $K=90>K_{cr}$ (encircles $-1$)', 'Interpreter', 'latex', 'FontSize', 13)
grid on
fprintf('K=30: closed-loop RHP poles = %d; K=90: closed-loop RHP poles = %d\n', ...
    sum(real(pole(feedback(tf(30,den3),1)))>0), ...
    sum(real(pole(feedback(tf(90,den3),1)))>0))
```

```
K=30: closed-loop RHP poles = 0; K=90: closed-loop RHP poles = 2
```



Try it yourself

- Sweep K between 30 and 90 and find (by counting encirclements of -1) the gain where the closed loop first goes unstable – it is $K_{cr}=60$.
- For the unstable-open-loop plant, check that closing the loop is actually stable by inspecting `pole(feedback(G3,1))`.